

U-Net–Initialized GAC: Testing the Two-Stage Hybrid

Does a classical level set seeded by a real detector beat the detector’s own boundary?

CS269 Project — Deformable Models

May 29, 2026

1 Question

The boundary-metrics report closed with an explicit, untested conjecture: the two-stage pipeline “U-Net to detect, GAC oracle-style to refine boundary” was a no-op on IoU when it was first tried, but *might* win on *boundary* metrics, because oracle-initialized GAC placed edges more accurately than the per-pixel U-Net even when its IoU was worse. This mini report runs that exact pipeline and scores it on the same boundary suite.

Method (`gac_unet`). For each test frame: the trained U-Net (`models/unet.pt` on FLAME-3, `models/flame1_unet.pt` on FLAME-1) predicts a 0/255 mask; that mask seeds `morphological_geodesic_active_contour` via `run_gac(frame, GACConfig())`. The *only* change versus the `gac_oracle` row is the initialization source: a real detector’s prediction instead of the ground-truth mask. GAC’s edge feature (`inverse_gaussian_gradient` of the R–G fire energy) and all hyperparameters are unchanged, so the experiment isolates the init source. The method is scored on the same deep test split (94 FLAME-3 / 301 FLAME-1 frames) as the `unet` row, making the U-Net-`vs-gac_unet` delta a clean apples-to-apples comparison. When the U-Net predicts nothing, the level set has no seed and the prediction is empty (scored like any other empty mask).

2 Results

Dataset	Method	n	IoU	Dice	BF @2	AvgSurf (px)	PolyChf (px)	PolyHaus (px)
FLAME-3	GAC (oracle)	592	0.365	0.499	0.409	16.5	17.3	131
FLAME-3	U-Net	94	0.147	0.207	0.157	54.9	53.5	204
FLAME-3	gac_unet	94	0.150	0.211	0.139	48.1	48.7	206
FLAME-3	DALS	94	0.122	0.182	0.131	48.9	47.1	194
FLAME-1	U-Net	301	0.772	0.870	0.757	4.60	6.53	141
FLAME-1	DALS	301	0.747	0.854	0.685	7.73	8.12	212
FLAME-1	gac_unet	301	0.646	0.784	0.507	6.43	7.62	151
FLAME-1	GAC (oracle)	2001	0.592	0.738	0.635	2.92	4.16	71

Table 1: `gac_unet` against its natural comparators. Higher is better for IoU/Dice/BF; lower is better for the distance columns (AvgSurf, PolyChamfer, PolyHausdorff, all px). Bold = best in column on that dataset. `gac_unet` and `unet` share the same test split (94/301); `gac_oracle` is the full classical sweep (592/2001) and is included for reference only—its mean is over a different denominator.

3 Verdict: the conjecture fails

The boundary report’s hypothesis is **not** supported once GAC is seeded by a real detector rather than the oracle. The boundary advantage it reported belonged to the *oracle* initialization, and it does not survive the substitution.

FLAME-3: near-wash, a sliver of boundary gain. The original “no-op on IoU” holds ($0.147 \rightarrow 0.150$). GAC *does* pull the U-Net boundary modestly toward the GT—AvgSurf $54.9 \rightarrow 48.1$ px, PolyChamfer $53.5 \rightarrow 48.7$ px ($\sim 10\%$ better)—but BF@2px actually *drops* ($0.157 \rightarrow 0.139$), and every number remains far from oracle-GAC’s 16.5/17.3 px. The U-Net’s seed on this hidden-thermal-fire task is too sparse and too often wrong for the R–G energy to refine into a good boundary. A small, equivocal improvement, not the predicted win.

FLAME-1: a clear regression. On visible flame, seeding GAC with the U-Net *degrades every metric* relative to the U-Net alone—region *and* boundary: IoU $0.772 \rightarrow 0.646$, AvgSurf $4.60 \rightarrow 6.43$ px, PolyChamfer $6.53 \rightarrow 7.62$ px. The classical edge energy drags the already-accurate U-Net boundary *off* the true flame edge. The clean “deformable model tracks edges better” result from the boundary report was entirely an artifact of the oracle (GT) seed: oracle-GAC’s PolyChamfer is 4.16 px, but `gac_unet`’s is 7.62 px—worse than U-Net’s 6.53.

Takeaway. The proposed two-stage hybrid does not beat the U-Net’s own boundary. GAC’s edge-localization strength in this project is inseparable from knowing the fire’s location in advance (the oracle). Handed a real detector’s output, the classical level set has nothing better than R–G redness to lock onto, so on the hidden-fire task it barely moves the boundary, and on the visible-flame task it makes a good boundary worse. If a boundary-accurate fire segmenter is the goal, the productive direction is a detector with an explicit edge-aware objective (e.g. DALs’s trained level set, or a boundary loss on the U-Net)—not a classical refinement step bolted on after detection.

4 Reproducing

```
python make_boundary_table.py    # gac_unet rows now included for both datasets
```

Per-frame CSVs: `results/gac_unet_boundary_per_frame.csv` (FLAME-3, 94) and `results/flame1_gac_unet_boundary_per` (FLAME-1, 301); summary rows in `results/boundary_summary.csv`. Wiring: `predict.gac_unet` in `make_boundary_table.py` `flame/unet_seed`.py generalized to select the per-dataset checkpoint.